

SignBridge - Gamified sign learning

Project Proposal

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CE–Cybersecurity

CMPE-112L PFDS

Project Overview

SignBridge is a standalone Python desktop application that makes American Sign Language (ASL) education accessible, engaging, and bilingual for Pakistani learners. It combines a real-time AI hand-sign recogniser, a complete gamification engine, and Urdu-language Text-to-Speech support — all within a professionally styled PyQt5 interface.

Course	CMPE-112L — Programming Fundamentals & Data Structures
Department	Computer Engineering – Cybersecurity
Platform	Windows / Linux Desktop (Python 3.10+)
Interface	PyQt5 5.15+ with full QSS dark theme
Database	SQLite — auto-created on first launch
Submission Date	29 June 2026

Objectives

- Provide a structured ASL curriculum: alphabet (A–Z), common words, and numbers.
- Integrate real-time AI hand-sign recognition using TensorFlow + OpenCV + MediaPipe.
- Sustain learner motivation via XP points, level tiers, streaks, badges & leaderboard.
- Deliver bilingual TTS output in English and Urdu (gTTS + deep-translator).
- Present a robust, scalable desktop UI using PyQt5 with QSS stylesheet support.

Core Features

10 Application Screens Login, Home, Learn, Practice, Quiz, Speak, Stats, Leaderboard, Profile, Settings	Real-Time AI Recognition TensorFlow SavedModel + MediaPipe landmark extraction; 1.5s hold-to-confirm logic
Gamification Engine XP, levels, daily streaks, 7 achievement badges (First Sign, Quiz Ace, Centurion...)	Spaced Repetition Quiz 10-question sessions (FLASHCARD + SIGN_IT); SRS weights prioritise difficult signs
Bilingual TTS (Urdu + English) Sign letters to assemble words; auto-translate to Urdu; gTTS playback with offline fallback	Live Stats Dashboard Matplotlib charts embedded via Qt5Agg: 7-day activity, accuracy trend, curriculum donut

Module

Component	Technology & Role
GUI	PyQt5 5.15+ — QMainWindow, QStackedWidget, QSS dark stylesheet
AI / Vision	TensorFlow 2.13+ + OpenCV — real-time hand-sign classification
Gesture	MediaPipe — 21-landmark hand extraction for model input
Database	SQLite + bcrypt — accounts, progress, quiz history
Charts	Matplotlib (Qt5Agg) — embedded stat visualisations
TTS / NLP	gTTS + deep-translator — English/Urdu speech synthesis
Audio	pygame + pyttsx3 — online & offline audio playback

Modular Architecture

The codebase is divided into four top-level packages with clear separation of concerns:

- auth/ — Login & Registration screens with bcrypt password verification
- ui/ — 9 PyQt5 screen classes; each a self-contained QWidget subclass
- modules/ — Business logic: gamification, quiz engine, SRS, ASL detector, TTS
- database/ — DatabaseManager wrapper; schema.sql; idempotent lesson seeder

UI Design Language

Dark professional palette with teal accent (#00d4aa), sidebar navigation (fixed 200 px QFrame), and QStackedWidget content area. QSS stylesheets applied globally; no third-party theme libraries.

Future Scope

Executable Packaging	PyInstaller .exe / .app — removes Python dependency for end users
Web / Mobile Port	React + FastAPI (web); Flutter (mobile) — browser / device camera
Multi-Language TTS	Extend to Punjabi, Sindhi, Pashto for broader Pakistani reach
Classroom Mode	Teacher dashboard: class progress tracking, lesson assignment, attendance export

Real-World Impact

- Deaf & Hard-of-Hearing Communities: trains hearing individuals to communicate with Deaf peers.
- Healthcare: rapid ASL skill-building for nurses, doctors, and emergency responders.
- Schools: digital inclusion tool for Pakistani special education departments.
- Corporate D&I: employee onboarding tool for Deaf-inclusive workplaces.